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Candidate Name :

RUTVIK AVINASH BARBHAI

BTech_Pharmacy and IMS



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10-12-2003



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Instructions

General Instructions:

1. The clock will be set at the server. The countdown timer in the top right corner of screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You will not be required to end or submit your examination.
2. The Question Palette displayed on the right side of screen will show the status of each question using one of the following symbols:
 - 1 You have not visited the question yet.
 - 2 You have not answered the question.
 - 3 You have answered the question.
 - 4 You have marked this question for review without answering.
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RUTVIK AVINASH BARBHAI

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English



Chemistry MCQ



Physics MCQ



Mathematics MCQ



Chemistry NAT



Physics NAT



Mathematics NAT



**RUTVIK AVINASH
BARBHAI**

Time Left :11:25

Sections

English



Question Type: Multiple Choice Question

Question No. 1

The research _____ for three years at the institute and we are expecting some positive results soon.

- ☐ A. had been going on
- ☐ B. is going on
- ☐ C. was going on
- ☐ D. has been going on

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English



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Physics MCQ



Mathematics MCQ



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**RUTVIK AVINASH
BARBHAI**

Sections

Time Left :11:25

English



Question Type: Multiple Choice Question

Question. 2

The passengers are not allowed to use cell phones _____ airplanes.

- ☐ A. across
- ☐ B. on
- ☐ C. in
- ☐ D. over

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English



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**RUTVIK AVINASH
BARBHAI**

Sections

Time Left :11:25

English



Question Type: Multiple Choice Question

Question No. 3

The Lawyer has plenty of _____

- ☐ A. Criminals
- ☐ B. Buyers
- ☐ C. Customers
- ☐ D .Clients

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English 

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**RUTVIK AVINASH
BARBHAI**

Time Left :11:25

Sections

English 

Question Type: Multiple Choice Question

Question No. 4

Identify the grammatically incorrect sentence among the following :

- ☐ A. Despite the differences in their age, they were close friends
- ☐ B. Despite of the differences in their age, they were close friends
- ☐ C. Despite the differences in their ages, they were close friends
- ☐ D. In spite the differences in their ages , they were close friends

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**RUTVIK AVINASH
BARBHAI**

Sections

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English



Question Type: Multiple Choice Question

Question No. 5

This exam is her best chance to prove her credentials. Everyone ____ and hoping for best

- ☐ A. is keeping their fingers crossed
- ☐ B. Washing the dirty linen
- ☐ C. Keeping their fingers closed
- ☐ D. Washing their hands off

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**RUTVIK AVINASH
BARBHAI**

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Sections

English 

Question Type: Multiple Choice Question

Question No. 6

Being a part of the institute ____ the past three years has certainly been ____ unique opportunity.

- ☐ A. since ; a
- ☐ B. for ; an
- ☐ C. for ; a
- ☐ D. since ; an

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English



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**RUTVIK AVINASH
BARBHAI**

Sections

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English



Question Type: Multiple Choice Question

Question No. 7

Either the method used by the student or the techniques introduced by the teacher _____ a re-examination before we move ahead.

- ☐ A. need
- ☐ B. is in need of
- ☐ C. needs
- ☐ D. has needed

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English 

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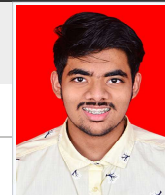
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BARBHAI**

Sections

Time Left :11:25

English 

Question Type: Multiple Choice Question

Question No. 8

I have brought the book. Its _____ ! Ravi said assertively to all the boys present

- ☐ A. mine
- ☐ B. my
- ☐ C. me
- ☐ D. myself

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**RUTVIK AVINASH
BARBHAI**

Sections

Time Left :11:25

English



Question Type: Multiple Choice Question

Question No. 9

The project is good , but there is _____ missing to make it an excellent work, “the engineer commented.

- ☐ A. Everything
- ☐ B. Anything
- ☐ C. Something
- ☐ D. Nothing

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**RUTVIK AVINASH
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Sections

English 

Question Type: Multiple Choice Question

Question No. 10

In the given words select the antonym of the given word: Barren

- ☐ A. Fertile
- ☐ B. Rich
- ☐ C. Prosperous
- ☐ D. Positive

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Sections

Chemistry MCQ 

Question Type: Multiple Choice Question

Question No. 1

Arrange the following in increasing order of the volume (in L) occupied by them at STP

(i) 1.5 moles of CO_2

(ii) 14 g of N_2

(iii) 1021 molecules of oxygen

- ☐ A. (iii) < (ii) < (i)
- ☐ B. (i) < (ii) < (iii)
- ☐ C. (ii) < (i) < (iii)
- ☐ D. (i) < (iii) < (ii)

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Sections

Chemistry MCQ 

Question Type: Multiple Choice Question

Question No. 2

According to VSEPR theory, the molecular shapes of XeF_4 , XeO_4 ,

XeO_2 , F_2 and XeOF_4 respectively are :

- ☐ square planar, square planar, see-saw, square pyramidal.
- ☐ square planar, tetrahedral, trigonal bipyramidal, octahedral.
- ☐ square planar, tetrahedral, see-saw, square pyramidal.
- ☐ octahedral, tetrahedral, trigonal bipyramidal, octahedral.

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Chemistry MCQ



Question Type: Multiple Choice Question

Question No. 3

If gas absorbs 150 J of heat and expands by 450 cm³ against a constant pressure of $2 \times 10^5 \text{ Nm}^{-2}$, then change in internal energy is?

- ☐ -60J
- ☐ 120J
- ☐ 240J
- ☐ -240J

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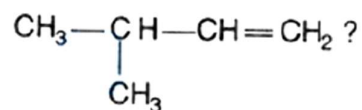
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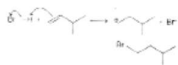
Question Type: Multiple Choice Question

Question No. 4

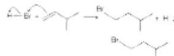
What is the first step and the final product formed in the reaction of H – Br with



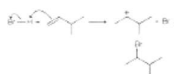
A.



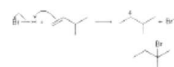
B.



C.



D.



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Chemistry MCQ



Question Type: Multiple Choice Question

Question No. 5

In case of positive deviation from Raoult's law, the intermolecular attractive forces between the solute-solvent molecules as compared to those between the solute-solute and solvent-solvent molecules are

- ☐ Weaker
- ☐ Stronger
- ☐ Same
- ☐ Independent of intermolecular force between solute-solvent molecules.

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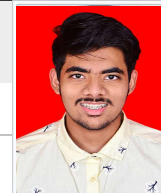
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Chemistry MCQ



Question Type: Multiple Choice Question

Question No. 6

Mond's process is used for refining _____. Whereas van Arkel method is used for refining _____.

- ☐ A) lead, zirconium
- ☐ B) zirconium, nickel
- ☐ C) nickel, lead
- ☐ D) nickel, titanium

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Chemistry MCQ



Question Type: Multiple Choice Question

Question No. 7

Select the correct statement.

- A. $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic whereas $[\text{Ni}(\text{CO})_4]$ is paramagnetic.
- B. $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$ both are diamagnetic.
- C. $[\text{Ni}(\text{CN})_4]^{2-}$ is sp^3 hybridised and square planar in shape whereas $[\text{Ni}(\text{CO})_4]$ is dsp^2 hybridised and tetrahedral in shape.
- D. $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic and dsp^2 hybridised whereas $[\text{Ni}(\text{CO})_4]$ is diamagnetic and sp^3 hybridised.

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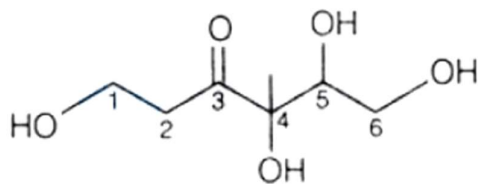
Chemistry MCQ



Question Type: Multiple Choice Question

Question No. 8

Alcoholic group of which position in the given molecule reacts fastest with the Lucas' reagent?



- A. 1
- B. 4
- C. 5
- D. 6

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English



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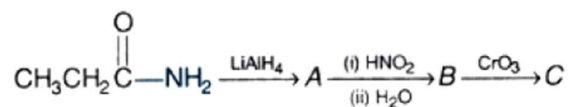
Chemistry MCQ



Question Type: Multiple Choice Question

Question No. 9

In the following sequence of reactions, the compound C formed would be:



- A. 2-propanol
- B. propanol
- C. propanoic acid
- D. propanal

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English



Chemistry MCQ



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Sections

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Chemistry MCQ



Question Type: Multiple Choice Question

Question No. 10

Polystyrene is a _____ polymer whereas dacron is a _____ polymer?

- ☐ step growth, chain growth
- ☐ chain growth, step growth
- ☐ condensation, addition
- ☐ thermoplastic, thermosetting

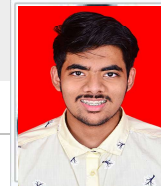
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Sections

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Physics MCQ 

Question Type: Multiple Choice Question

Question No. 1

The time period of a mass suspended from a spring is T . If the spring is cut into three equal parts and connected in parallel. The same mass is suspended from these parallel springs, then the new time period of the mass will be :

- ☐ $T/4$
- ☐ T
- ☐ $T/3$
- ☐ $3T$

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Physics MCQ 

Question Type: Multiple Choice Question

Question No. 2

The work done by all the forces (external and internal) on a system is equal to change in

- A. Total energy
- B. Kinetic energy
- C. Potential energy
- D. None of These

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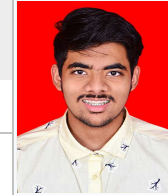
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Sections

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Physics MCQ 

Question Type: Multiple Choice Question

Question No. 3

If the radius of a planet is three times the radius of the earth. Both have same mass-densities. v_P and v_E are the escape velocities of the planet and the earth respectively, then

- A. $v_P = 1.5v_E$
- B. $v_P = 3v_E$
- C. $v_E = 2v_P$
- D. $v_P = 2v_E$

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**RUTVIK AVINASH
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Sections

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Physics MCQ 

Question Type: Multiple Choice Question

Question No. 4

At which temperature, magnitude of $^{\circ}\text{C}$ and $^{\circ}\text{F}$ are equal?

- A. 273
- B. 40
- C. -40
- D. -273

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Physics MCQ



Question Type: Multiple Choice Question

Question No. 5

The potential at a point at a distance r from the centre of an electric dipole is proportional to :

- A. $1/r$
- B. $1/r^2$
- C. $1/r^3$
- D. r^2

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Physics MCQ



Question Type: Multiple Choice Question

Question No. 6

The electric resistance of a wire is R . If the length of the wire is increased to doubled by stretching it, then the new resistance of the wire is

A. $2R$

B. $4R$

C. R

D. $16R$

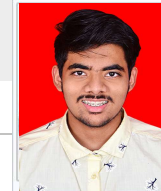
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Physics MCQ 

Question Type: Multiple Choice Question

Question No. 7

The angle of a prism is A . One of its refracting surface is silvered. If light rays: falling at an angle of incidence $2A$ on the first surface returns back through the same path after reflection from silvered surface. The refractive index μ of the prism is

- ☐ A. $2\sin A$
- ☐ B. $2\cos A$
- ☐ C. $1/2\cos A$
- ☐ D. $\tan A$

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Physics MCQ

Question Type: Multiple Choice Question

Question No. 8

An electron, helium ion (He^{++}) and proton having the same kinetic energy. The relation between their respective de-Broglie wavelengths λ_e , $\lambda_{\text{He}^{++}}$ and λ_p is

- A. $\lambda_e > \lambda_p > \lambda_{\text{He}^{++}}$
- B. $\lambda_e > \lambda_{\text{He}^{++}} > \lambda_p$
- C. $\lambda_e < \lambda_p < \lambda_{\text{He}^{++}}$
- D. $\lambda_e < \lambda_{\text{He}^{++}} = \lambda_p$

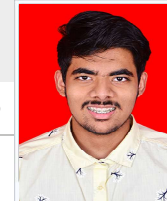
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Question Type: Multiple Choice Question

Question No. 9

A particle of charge q and mass m is moving with a velocity towards a large screen placed in $Y-Z$ -plane placed at a distance d . If there is a magnetic field the maximum value of v for which the particle will not strike the screen is

- A. $\frac{qdB_0}{m}$
 B. $\frac{qdB_0}{2m}$
 C. $\frac{2qqdB_0}{m}$
 D. $\frac{qdB_0}{3m}$

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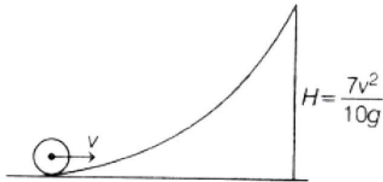
Question Type: Multiple Choice Question

Question No. 10

A small object of uniform density rolls up a curved surface with an initial velocity v .

It reaches upto a maximum height of $\frac{7v^2}{10g}$ with respect to initial position.

Then the object is



- A. ring
- B. solid sphere
- C. hollow sphere
- D. disc

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**RUTVIK AVINASH
BARBHAI**

Sections

Time Left : 11:25

Mathematics MCQ

Question Type: Multiple Choice Question

Question No. 1

1. If $|z + \frac{2}{z}| = 2$, then the minimum value of $|z|$ is

- ☐ A. $1 + \sqrt{2}$
- ☐ B. $1 + 2\sqrt{2}$
- ☐ C. $3\sqrt{3} + 1$
- ☐ D. $1 - \sqrt{3}$

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Mathematics MCQ

Question Type: Multiple Choice Question

Question No. 2

If α , β and γ are in AP and $\tan^{-1} \alpha$, $\tan^{-1} \beta$ and $\tan^{-1} \gamma$ are also in AP, then

- ☐ A. $\alpha - \beta - \gamma = 0$
- ☐ B. $\alpha = \beta = \gamma$
- ☐ C. $\alpha + \beta = \gamma$
- ☐ D. $2\alpha = 3\beta = \gamma$

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Mathematics MCQ

Question Type: Multiple Choice Question

Question No. 3

The coefficient of the middle term in the binomial expansion in powers of x of $(1 + \alpha x)^4$ and $(1 - \alpha x)^6$ is the same, if α is equal to

- ☐ A. $-5/3$
- ☐ B. $10/3$
- ☐ C. $-3/10$
- ☐ D. $3/5$

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Mathematics MCQ



Question Type: Multiple Choice Question

Question No. 4

The differential equation of all circles passing through the origin and having their centre on the X-axis is

A. $x^2 = y^2 + xy \frac{dy}{dx}$

B. $x^2 = y^2 + 3xy \frac{dy}{dx}$

C. $y^2 = x^2 + 2xy \frac{dy}{dx}$

D. $y^2 = x^2 - 2xy \frac{dy}{dx}$

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Mathematics MCQ

Question Type: Multiple Choice Question

Question No. 5

The value of $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x}$

- ☐ A. 3
- ☐ B. 2
- ☐ C. 1
- ☐ D. -1

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Sections

Mathematics MCQ

Question Type: Multiple Choice Question

Question No. 6

If $\sum_{i=1}^{10} (x_i - 3) = 7$ and $\sum_{i=1}^{10} (x_i - 3)^2 = 27$, then the standard deviation of the 10 items

- ☐ A. 2.547
- ☐ B. 1.87
- ☐ C. 14.86
- ☐ D. 1.486

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Question Type: Multiple Choice Question

Question No. 7

If $A = \{1,2\}$, $B = \{1,2,3,4\}$, $C = \{5,6\}$ and $D = \{5,6,7,8\}$, then state which of the following is true?

- ☐ A. $(A \times C) \subset (B \times D)$
- ☐ B. $A \times B \subset C \times D$
- ☐ C. $(A \times B) \subset (A \times D)$
- ☐ D. $A \times C \subset B \times D$

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Question Type: Multiple Choice Question

Question No. 8

If A and $n \times n$ non-singular matrix such that $AA^T = A^{-1}A$ and $B = A^{-1}A^T$, then BB' is equal to

- ☐ A. $1 + B$
- ☐ B. 1
- ☐ C. B^{-1}
- ☐ D. $(B^{-1})'$

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Question Type: Multiple Choice Question

Question No. 9

Let $f(x) = \sqrt{1+x^2}$, then

- ☐ A. $f(xy) = f(x) \cdot f(y)$
- ☐ B. $f(xy) \geq f(x) \cdot f(y)$
- ☐ C. $f(x,y) \leq f(x) \cdot f(y)$
- ☐ D. $f(xy) = f(x) - f(y)$

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Question Type: Multiple Choice Question

Question No. 10

For what value of θ lying between 0 and π which satisfy inequality $\sin\theta \cos^3 \theta > \sin^3 \theta \cos\theta$

- ☐ A. $\theta \in (\pi/4, \pi/2)$
- ☐ B. $\theta \in (0, \pi/4)$
- ☐ C. $\theta \in (0, \pi/2)$
- ☐ D. None of these

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Question Type: Multiple Choice Question

Question No. 11

If $\int f(x)dx = g(x)$, then $\int x^9 \cdot f(x^5) dx$ is equal to

- ☐ A. $1/5 (x^5 g(x^9) - 4 \int g(x)dx + C$
- ☐ B. $1/5 [x^9 g(x^5) - 1/5 \int x^4 g(x^5) dx + C$
- ☐ C. $1/5 [g(x^9) + \int g(x^5) dx]$
- ☐ D. $\frac{x^6}{5} g(x^5) - \int x^4 g(x^5) dx + C$

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Question Type: Multiple Choice Question

Question No. 12

The area of the region bounded by the curves $y = |x - 2|$, $x = 1$, $x = 3$ and the x-axis is

- ☐ A. 1 sq unit
- ☐ B. 2 sq units
- ☐ C. 3 sq units
- ☐ D. 4 sq units

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Question Type: Multiple Choice Question

Question No. 13

The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is

- ☐ A. $\pi/2 + 4/3$
- ☐ B. $\pi/2 - 1/2$
- ☐ C. $\pi/4 + 2/3$
- ☐ D. $\pi/2 + 2/3$

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Question Type: Multiple Choice Question

Question No. 14

If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{2k} = \frac{y-3}{2} = \frac{z-5}{1}$ are coplanar, then k can have

- ☐ A. exactly one value, $k = 1/2$
- ☐ B. exactly one value, $k = 1/4$
- ☐ C. exactly two values, $k = 1/2, -3/2$
- ☐ D. any value

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Question Type: Multiple Choice Question

Question No. 15

A multiple choice examination has 5 questions. Each question has 4 alternative answer of which exactly one is correct. The probability that a student will get 4 or more correct answer just by guessing is

- ☐ A. $1/4^5$
- ☐ B. $(3/4)^4$
- ☐ C. $1/4^3$
- ☐ D. $3/4^5$

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Question Type: Numerical Answer Type

Question No. 1

From the following molar conductivities at infinite dilution, Λ_m° (in $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$) for NH_4OH is

$$\Lambda_m^\circ \text{ for } \text{Ba}(\text{OH})_2 = 446.8 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\Lambda_m^\circ \text{ for } \text{BaCl}_2 = 241.6 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\Lambda_m^\circ \text{ for } \text{NH}_4\text{Cl} = 130 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

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Question Type: Numerical Answer Type

Question No. 2

For the reaction, $\text{N}_2\text{O}_5(\text{g}) \rightarrow 2\text{NO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g})$ the value of the rate of disappearance of N_2O_5 is given as $5.15 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$. The rate of formation of NO_2 is.

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Question Type: Numerical Answer Type

Question No. 3

The number of oxygen atoms that are directly attached to one chromium in dichromate ion are

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Question Type: Numerical Answer Type

Question No. 4

The number of oxygen atoms that are directly attached to one chromium in dichromate ion are

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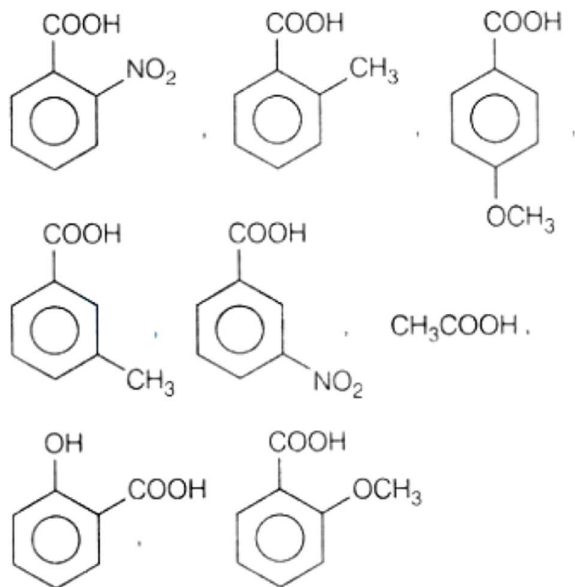
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Question Type: Numerical Answer Type

Question No. 5

Out of the given compounds, the number of compounds are weaker acids than benzoic acid are :



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Question Type: Numerical Answer Type

Question No. 1

A wire in the form of semi-circle of radius r rotates about its diameter with angular velocity ω in a magnetic field B . The axis of rotation is perpendicular to the field. The total resistance of the circuit is R . If the mean power generated per period of rotation is $\frac{(B\pi r^2(\omega)^2)}{xR}$, then the value of x is

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Question Type: Numerical Answer Type

Question No. 2

The stress that has to be applied to the ends of a steel wire of length 20 cm to keep its length constant, when its temperature is raised by 100°C is $2.2 \times 10^x \text{ Pa}$. The value of x is

(Given $Y = 2 \times 10^{11} \text{ Nm}^{-2}$,

$\alpha = 1.1 \times 10^{-5} \text{ C}^{-1}$)

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Question Type: Numerical Answer Type

Question No. 3

A vessel contains oil (Density = 0.8 g/cm^3) over mercury (density = 13.6 g/cm^3). A homogeneous sphere floats with half of its volume immersed in mercury and the other half in oil. The density of the material of the sphere is $x \text{ g/cm}^3$. The value of x is

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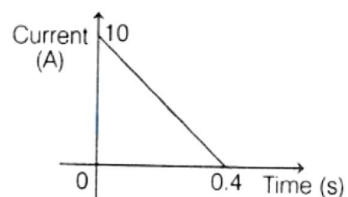
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Question Type: Numerical Answer Type

Question No. 4

In a coil of resistance 150Ω , a current is induced by changing the magnetic flux through it as shown by figure. The magnitude of flux through the coil is Wb.



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Question Type: Numerical Answer Type

Question No. 5

The electric field of light wave is given is

$$E = 10^3 \cos\left(\frac{2\pi x}{5 \times 10^{-7}} - 2\pi \times 6 \times 10^{14} t\right) \hat{j} \text{ N/C}.$$

This light falls on a metal plate of work function 1.5 eV. The stopping potential of the photoelectron is V.

$$(\text{Energy of photon} = \frac{1240}{\lambda(\text{in nm})} \text{ eV}).$$

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Question Type: Numerical Answer Type

Question No. 1

The number of 4 letter word (with or without meaning) that can be formed from the eleven letters of the word 'EXAMINATION is

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Question Type: Numerical Answer Type

Question No. 2

$\lim_{x \rightarrow 2} \frac{x^2 + 2^2 - x - 5}{2^{-x/2} - 2^{1-x}}$ is equal to _____

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Question Type: Numerical Answer Type

Question No. 3

If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and vectors $(1, a, a^2)$, $(1, b, b^2)$ and $(1, c, c^2)$ are non-coplanar, then the product abc is _____

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Question Type: Numerical Answer Type

Question No. 4

If the foot of the perpendicular drawn from the point $(0, 2, 1)$ on a line passing through $(\alpha, 5, 1)$ is $(5/3, 7/3, 15/i)$, then α is equal to _____

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Question Type: Numerical Answer Type

Question No. 5

If the probability that a randomly chosen 6-digit number formed by using digits 1 and 8 only is a multiple of 21 is p , then $96p$ is equal to

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